Module 4: Emergencies & Master
EmergencyStatus
PENDING
RESOLVED
UNRESOLVED
EmergencyManager
handleEmergency(e: Emergency): void
Main
runScenarios(): void
Handles
Emergency
id: String
location: Location
status: EmergencyStatus
resolve(): void
OxygenLeakEmergency
FireEmergency
RoverBreakdownEmergency
MedicalEmergency
Runs
Module 3: Missions & Simulation
MissionState
CREATED
ASSIGNED
IN_PROGRESS
COMPLETED
FAILED
Simulation
clock: int
advanceTime(units: int): void
MarsColony
activeMission: Mission
submitMission(m: Mission): void
generateReport(): void
Association
Mission
id: String
state: MissionState
duration: int
elapsedTime: int
requiredAgents: List<Agent>
advanceTime(): void
Aggregation
ExplorationMission
RepairMission
ResearchMission
RescueMission
Module 2: Facilities & Resour
uses
(Member 2)
ResourceType
OXYGEN
WATER
FOOD
FUEL
BATTERY_POWER
MEDICAL_SUPPLIES
SPARE_PARTS
ResourceManager
facilities: List<Facility>
addFacility(facility: Facility): void
getFacilities(): List<Facility>
getTotalResource(type: ResourceType): int
getAllResourceTotals(): Map<ResourceType, Integer>
validateRequest(request: Map<ResourceType, Integer>): void
hasEnoughResources(request: Map<ResourceType, Integer>): boolean
consumeResources(request: Map<ResourceType, Integer>): boolean
getShortageReason(request: Map<ResourceType, Integer>): String
Aggregation
manages
Facility
facilityId: String
location: String
capacity: int
operational: boolean
equipment: ArrayList<Equipment>
resources: Map<ResourceType, Integer>
getFacilityId(): String
getLocation(): String
getCapacity(): int
isOperational(): boolean
setOperational(operational: boolean): void
addEquipment(item: Equipment): void
getEquipment(): List<Equipment>
hasWorkingEquipment(capability: String): boolean
getResourceQuantity(type: ResourceType): int
getResources(): Map<ResourceType, Integer>
addResource(type: ResourceType, quantity: int): void
consumeResource(type: ResourceType, quantity: int): boolean
loseLocalResource(type: ResourceType, quantity: int): int
located at
Aggregation
occurs at
Module 1: Agents & Environment
Environment
condition: EnvironmentCondition
getCondition(): EnvironmentCondition
allowsSurfaceOperations(): boolean
allowsRemoteCommunication(): boolean
Capability
SCIENTIFIC_RESEARCH
ENGINEERING_REPAIR
MEDICAL_CARE
SURFACE_EXPLORATION
AERIAL_RECON
EnvironmentCondition
NORMAL
DUST_STORM
COMMUNICATION_DISRUPTION
Agent
id: String
location: Location
isAssigned: boolean
isUsable(): boolean
canPerform(m: Mission): boolean
canRespondTo(e: Emergency): boolean
onTimeStep(): void
located at
Inheritance
Inheritance
Location
name: String
isSurface: boolean
CrewMember
name: String
health: int
fatigue: int
skills: List<Capability>
RoboticUnit
batteryLevel: int
isDamaged: boolean
Scientist
Engineer
MedicalOfficer
RepairRobot
Drone
ExplorationRover
Habitat
lifeSupport: LifeSupportSystem
isLifeSupportOperational(): boolean
stopOxygenCirculation(): void
stopAirFiltration(): void
repairLifeSupport(): void
applyOxygenLeak(quantity: int): int
Composition
LifeSupportSystem
oxygenCirculation: boolean
airFiltration: boolean
isOxygenCirculation(): boolean
isAirFiltration(): boolean
stopOxygenCirculation(): void
stopAirFiltration(): void
repair(): void
isOperational(): boolean
Laboratory
canConductResearch(): boolean
StorageFacility
PowerStation
getAvailablePower(): int
hasSufficientPower(requiredPower: int): boolean
CommunicationStation
communicationEnabled: boolean
canCommunicate(): boolean
disableCommunication(): void
enableCommunication(): void
Equipment
equipmentId: String
name: String
capability: String
damaged: boolean
getEquipmentId(): String
getName(): String
getCapability(): String
isDamaged(): boolean
damage(): void
repair(): void
canPerform(requiredCapability: String): boolean